BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 2191

Retyped faithfully from the original board paper — same questions, no changes.

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ. 1. 1122 — Mathematics — (Inter Part - I) — (Session 2018-20 to 2021-23) — Paper (I) — (Objective)

Q. Time Allowed: 30 minutes — Maximum Marks: 20 — PAPER CODE 2191

Q1. Imaginary part of i/(1+i) is

(A) (B)

(C) (D)

Q2. Under what conditions on A and B, the statement A − B = A is true

(A) (B)

(C) (D)

Q3. If order of A^t is m×n, then order of matrix A is

(A) (B)

(C) (D)

Q4. A square matrix A is skew hermitian if

(A) (B)

(C) (D)

Q5. Which of the following equation has the roots 2 and 3

(A) (B)

(C) (D)

Q6. If the sum of the roots of the equation $ax^2 - 2x + 2a = 0$ is equal to their product, then the value of a is

(A) (B)

(C) (D)

Q7. Partial fractions of $(2x+3)/(x^2(x^2-1))$ will be of the form

(A) (B)

(C) (D)

Q8. A.M between a/2 and 2/a is

(A) (B)

(C) (D)

Q9. The sum of first "n" natural number is

(A) (B)

(C) (D)

Q10. $r!\ {}^nC_r =$

(A) (B)

(C) (D)

Q11. If A and B are independent events, then $P(A \cap B) =$

(A) (B)

(C) (D)

Q12. Sum of binomial coefficients in the expansion of $(a+x)^5$ is

(A) (B)

(C) (D)

Q13. The expansion $(2-3x)^{-2}$ is valid if

(A) (B)

(C) (D)

Q14. Angle between the hands of a watch at 6 O'clock is

(A) (B)

(C) (D)

Q15. $\tan(\theta - 270°) =$

(A) (B)

(C) (D)

Q16. Period of cos θ is

(A) (B)

(C) (D)

Q17. Formula for circum-radius is

(A) (B)

(C) (D)

Q18. Area of a triangle ABC =

(A) (B)

(C) (D)

Q19. $\tan^{-1} x + \cot^{-1} x =$

(A) (B)

(C) (D)

Q20. Reference angle for $2\sin x - 1 = 0$ is

(A) (B)

(C) (D)

Pen tick-marks on the scans are a student's working, not transcribed. Board typos kept verbatim: 'hermitian', 'Skew hermitian', 'the following equation has the roots'.

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QSection I. 1122 — Mathematics — (Inter Part - I) — (Session 2018-20 to 2021-23) — Paper (I) — (Subjective)

Q. Time Allowed: 2.30 hours — Maximum Marks: 80

Q. Section ---------- II — (10 × 3 = 30)

Q. Note: Attempt any three questions.